

Robot Impedance Iterative Learning With Sparse Online Gaussian Process

Yongping Pan , Senior Member, IEEE, Tian Shi , Member, IEEE, Wei Li ,
Bin Xu , and Choon Ki Ahn , Senior Member, IEEE

Abstract—Robot interaction control with variable impedance parameters may conform to task requirements during continuous interaction with dynamic environments. Iterative learning (IL) is effective to learn desired impedance parameters for robots under unknown environments, and Gaussian process (GP) is a nonparametric Bayesian approach that models complicated functions with provable confidence using limited data. In this paper, we propose an impedance IL method enhanced by a sparse online Gaussian process (SOGP) to speed up learning convergence and improve generalization. The SOGP for variable impedance modeling is updated in the same iteration by removing similar data points from previous iterations while learning impedance parameters in multiple iterations. The proposed IL-SOGP method is verified by high-fidelity simulations of a collaborative robot with 7 degrees of freedom based on the admittance control framework. It is shown that the proposed method accelerates iterative convergence and improves generalization compared to the classical IL-based impedance learning method.

Index Terms—Gaussian process (GP), impedance variation, iterative learning (IL), physical robot interaction, robot learning.

I. INTRODUCTION

WHILE robots are in direct contact with environments, compliance is required during interaction to avoid risky situations [1]. Compliance control is a major interaction control approach that allows robots to perform ideal characteristics similar to a mass-spring-damping model [2]. This can be achieved by shaping robot impedance, which introduces virtual robot compliance to ensure safe and natural interaction.

Manuscript received July 3, 2024; revised November 8, 2024; accepted January 13, 2025. This work was supported in part by the National Research Foundation of Korea (NRF) Grant Funded by the Korea Government (MSIT) (RS-2025-00555064). Recommended by Associate Editor Zengguang Hou. (Corresponding author: Yongping Pan.)

Citation: Y. Pan, T. Shi, W. Li, B. Xu, and C. K. Ahn, “Robot impedance iterative learning with sparse online Gaussian process,” *IEEE/CAA J. Autom. Sinica*, vol. 12, no. 11, pp. 2218–2227, Nov. 2025.

Y. Pan is with the School of Automation, Southeast University, Nanjing 210096, China, and also with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore 639798, Singapore (e-mail: eee-yppan@ntu.edu.sg).

T. Shi is with the School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou 510006, China (e-mail: shit23@alumni.sysu.edu.cn).

W. Li is with the School of Software Engineering, Sun Yat-sen University, Guangzhou 510006, China (e-mail: liwei363@mail2.sysu.edu.cn).

B. Xu is with the School of Automation, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: binxu@nwpu.edu.cn).

C. K. Ahn is with the School of Electrical Engineering, Korea University, Seoul 02841, South Korea (e-mail: hironaka@korea.ac.kr).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/JAS.2025.125195

Compliance control can be grouped into force control-based impedance control and position control-based impedance control [3]. While the former is called impedance control directly, the latter is usually called admittance control. In many situations, compliance control with fixed impedance parameters does not meet task requirements [4], and robots must change impedance during continuous interaction with environments. Yet, efficiently learning and adjusting robot impedance to adapt to dynamic environments is still challenging.

Iterative learning (IL) was first developed for the accurate tracking of repetitive control processes with a wide range of applications [5]. It is also effective in impedance learning for robots under unknown environments, where desired impedance parameters are iterated during robot-environment interaction to minimize an objective function so as to achieve the desired robot behavior [6]. Nevertheless, IL has two major drawbacks: First, it needs to be re-iterated if the initial position or desired trajectory is changed [7]; second, there is no suitable way to guide the learning rate setting for some methods, making it difficult to ensure the convergence rate [6]. Therefore, impedance IL has weak generalization since the learned variable impedance knowledge is not applicable to different tasks or environments, which significantly restricts its practicality. Although IL can be combined with neural networks (NNs) to make it suitable for multiple tasks, it requires rich training data [8]. Gaussian process (GP) is a nonparametric Bayesian approach with modeling flexibility [9] and uncertainty estimation [10] and has the ability to model complicated functions using only a limited number of data [11]. Furthermore, sparse online GP (SOGP) has the online learning ability to update the GP model with arriving time-series data [12]–[14], so it is promising in robotic applications.

This study applies a GP model to better exploit the knowledge during impedance IL for adapting to new tasks. More specifically, we propose an IL-SOGP method to accelerate impedance learning convergence and improve generalization. The SOGP for variable impedance modeling is updated in the same iteration by removing similar data points from previous iterations while learning impedance parameters in multiple iterations. During the same iteration, impedance parameters learned at the current instant can immediately affect the next prediction, speeding up learning convergence. Besides, the variable impedance strategy from the proposed method is not task-dependent, such that it can effectively exploit the original knowledge when the initial robot position or task trajec-

tory is changed, and thus, the number of iterations can be shortened to accommodate new tasks. Our main contribution is proposing a learning-based variable impedance method that accelerates learning convergence while efficiently exploiting the learned knowledge, where a variable impedance strategy can be learned quickly by incorporating SOGP into IL.

Notations: $\mathbb{R}$, $\mathbb{R}^+$, $\mathbb{R}^n$, and $\mathbb{R}^{m \times n}$ denote the spaces of real numbers, positive real numbers, real n -vectors, and real $m \times n$ -matrices, respectively, $\|\mathbf{x}\|_2$ and $\|\mathbf{x}\|_\infty$ are the 2-norm and ∞ -norm of $\mathbf{x}$, respectively, $\mathcal{GP}(m(\mathbf{x}), \kappa(\mathbf{x}, \mathbf{x}'))$ is a GP model with a mean $m: \mathbb{R}^n \mapsto \mathbb{R}$ and a kernel $\kappa: \mathbb{R}^n \times \mathbb{R}^n \mapsto \mathbb{R}$, $\text{diag}\{x_1, x_2, \dots, x_n\}$ is a diagonal matrix with diagonal elements x_1 to x_n , A_{ij} is the ij th element of A , I is an identity matrix with a proper dimension, where $A \in \mathbb{R}^{n \times n}$, $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$, $x_i \in \mathbb{R}$, $i = 1$ to n , and n and m are positive integers.

II. RELATED WORK

Machine learning models are usually employed as function approximations to design control laws [15]–[17], but they are also frequently applied to learn impedance profiles online for robot interaction control [18]–[23]. In [18], a fuzzy inference system with the robot joint velocity and contact force as inputs was designed to adjust damping parameters, in which online training is done by a model reference learning mechanism based on the minimum jerk trajectory model. In [19], a feed-forward NN was applied to learn damping parameters, where NN weights are updated via a Cartesian velocity tracking error based on the minimum jerk trajectory model. The method of [19] was also utilized to learn inertia parameters in [20]. Yu *et al.* [21] applied a GP model to learn the human stiffness and desired trajectory under the assumption that the robot motion is slow and the human model contains only stiffness information. Li *et al.* [22] applied an GP model to represent the target impedance profile and employed reinforcement learning to get an optimal impedance regulation strategy, where stiffness and damping parameters are adjusted based on the contact force. Deng *et al.* [23] combined a SOGP model with gradient descent to enhance the learning efficiency of stiffness and damping parameters. The above methods are either limited by the forms of objective functions, resulting in restricted interaction effects, or have degraded generalization ability.

Iterative learning has also been applied to learn the desired impedance parameters for robots under unknown environments [6], [24]–[27], where an interaction task is executed repeatedly, and impedance parameters are updated at each iteration of the task execution. Li and Ge [6] proposed an impedance IL method with gradient-following and betterment schemes to obtain the target admittance model, where the desired interaction performance is achieved by minimizing an objective function relevant to robot joint position, joint velocity, and contact force. Based on [6], Wang *et al.* [24] further considered reference adaptation integrating trajectory parameterization. Yamawaki *et al.* [25] designed an impedance IL scheme to update damping parameters based on the minimization of the contact force. Kramberger *et al.* [26] introduced admittance-coupled dynamic movement primitives (DMPs)

for robots interacting with changing environments, where DMPs are updated by IL to generate the desired admittance profile. Li *et al.* [27] proposed an IL impedance control method for compliant rehabilitation robots. One major disadvantage of impedance IL is the high sensitivity to the changes of the initial position and desired trajectory, which results in weak generalization. Inspired by these works, this study aims to integrate a machine learning model (specifically, an SOGP model) into impedance IL to enhance its generalization.

III. THE PROPOSED METHOD

A. Iterative Impedance Learning Framework

Consider a robot-environment interaction problem, where the robot kinematics is described by

$$\mathbf{x} = \mathbf{f}(\mathbf{q}), \dot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}} \quad (1)$$

in which $\mathbf{x}(t) \in \mathbb{R}^6$ is the pose in the Cartesian space, whose orientation is represented by Euler angles, $\mathbf{q}(t) \in \mathbb{R}^n$ is the position in the joint space, $\mathbf{f}: \mathbb{R}^n \mapsto \mathbb{R}^6$ maps from the joint to Cartesian spaces, $\mathbf{J}(\mathbf{q}) \in \mathbb{R}^{6 \times n}$ is the Jacobian matrix, and n is the number of degrees of freedom (DoFs). The robot dynamics is described as follows:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{F}(\mathbf{q}, \dot{\mathbf{q}}) = \boldsymbol{\tau} - \mathbf{J}^T(\mathbf{q})\mathbf{f}_e \quad (2)$$

where $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{n \times n}$ is an inertia matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{n \times n}$ is a centripetal-Coriolis matrix, $\mathbf{F}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^n$ denotes a friction torque, $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^n$ is a gravitational torque, $\boldsymbol{\tau} \in \mathbb{R}^n$ is a control torque, and $\mathbf{f}_e(t) \in \mathbb{R}^6$ is an interaction force at the robot end-effector that can be measured by a force/torque sensor.

An unknown environment is represented by a mass-damping-spring model in the Cartesian space as follows:

$$\mathbf{M}_e(t)\ddot{\mathbf{x}} + \mathbf{B}_e(t)\dot{\mathbf{x}} + \mathbf{K}_e(t)\mathbf{x} = \mathbf{f}_e \quad (3)$$

in which $\mathbf{x}(t) \in \mathbb{R}^6$ denotes the pose of the environment, and $\mathbf{M}_e(t) \in \mathbb{R}^{6 \times 6}$, $\mathbf{B}_e(t) \in \mathbb{R}^{6 \times 6}$, and $\mathbf{K}_e(t) \in \mathbb{R}^{6 \times 6}$ are diagonal inertia, damping and stiffness matrices of the environment, respectively. Note that (3) is a classical and effective model to describe an unknown environment and is suitable for different environments as $\mathbf{M}_e(t)$, $\mathbf{B}_e(t)$, and $\mathbf{K}_e(t)$ can be changed in real-time [28]. Then, (3) is rewritten into a state-space form

$$\begin{cases} \dot{\boldsymbol{\chi}} = \mathbf{A}(t)\boldsymbol{\chi} + \mathbf{B}(t)\mathbf{f}_e \\ \mathbf{v} = \mathbf{C}\boldsymbol{\chi} \end{cases} \quad (4)$$

with $\mathbf{v}(t) \in \mathbb{R}^6$ and $\boldsymbol{\chi}(t) = [\mathbf{x}^T(t), \dot{\mathbf{x}}^T(t), (\int_0^t \mathbf{f}_e(\varsigma) d\varsigma)^T]^T \in \mathbb{R}^{18}$, where $\mathbf{A}(t) \in \mathbb{R}^{18 \times 18}$ and $\mathbf{B}(t) \in \mathbb{R}^{18 \times 6}$ are given by

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} \\ -\mathbf{M}_e^{-1}\mathbf{K}_e & -\mathbf{M}_e^{-1}\mathbf{B}_e & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \mathbf{0} \\ -\mathbf{M}_e^{-1} \\ \mathbf{I} \end{bmatrix}$$

and $\mathbf{C} \in \mathbb{R}^{6 \times 18}$ is determined based on the desired interaction effect. For (4), the contact force $\mathbf{f}_e(t)$ is iterated by

$$\mathbf{f}_e^k(t) = \mathbf{f}_e^{k-1}(t) + \alpha'[\dot{\mathbf{v}}_d(t) - \dot{\mathbf{v}}^k(t)] \quad (5)$$

with $\mathbf{v}^k(0) = \mathbf{v}_d(0)$, where k is the iteration number, $\mathbf{v}_d(t) \in \mathbb{R}^6$ is the desired output, and $\alpha' \in \mathbb{R}^+$ satisfies $\|I - \alpha' CB\|_\infty < 1$ with CB being a nonsingular matrix. Then, it follows from the generic betterment scheme [6] that $\mathbf{v}^k(t) \rightarrow \mathbf{v}_d(t)$ uniformly in $t \in [0, t_f]$ as $k \rightarrow \infty$, where $t_f \in \mathbb{R}^+$ is the iteration period.

The target impedance model is described by

$$M_d(t)(\ddot{\mathbf{x}}_r - \ddot{\mathbf{x}}_d) + B_d(t)(\dot{\mathbf{x}}_r - \dot{\mathbf{x}}_d) + K_d(t)(\mathbf{x}_r - \mathbf{x}_d) = \mathbf{f}_e \quad (6)$$

where $\mathbf{x}_d(t) \in \mathbb{R}^6$ and $\mathbf{x}_r(t) \in \mathbb{R}^6$ denote desired and reference poses of the end-effector, respectively, and $M_d(t) \in \mathbb{R}^{6 \times 6}$, $B_d(t) \in \mathbb{R}^{6 \times 6}$ and $K_d(t) \in \mathbb{R}^{6 \times 6}$ denote diagonal matrices of desired inertia, damping, and stiffness, respectively. Consider an objective function as follows:

$$\mathcal{L}(t) = \|\mathbf{v}(t) - \mathbf{v}_d(t)\|_2. \quad (7)$$

Note that $\mathcal{L}$ is task-dependent and not limited to the form in (7). Then, B_d and K_d can be iterated by

$$\begin{cases} B_d^k(t) = B_d^{k-1}(t) - \beta \left(\frac{\partial \mathbf{f}_e^k(t)}{\partial B_d^k(t)} \right)^T \left(\frac{\partial \mathcal{L}^k(t)}{\partial \mathbf{f}_e^k(t)} \right)^T \\ K_d^k(t) = K_d^{k-1}(t) - \beta \left(\frac{\partial \mathbf{f}_e^k(t)}{\partial K_d^k(t)} \right)^T \left(\frac{\partial \mathcal{L}^k(t)}{\partial \mathbf{f}_e^k(t)} \right)^T \end{cases} \quad (8)$$

in which $\beta \in \mathbb{R}^+$ denotes a learning rate. It is easy to obtain $(\frac{\partial \mathbf{f}_e^k(t)}{\partial K_d^k(t)})^T = \mathbf{e}^k(t)$ and $(\frac{\partial \mathbf{f}_e^k(t)}{\partial B_d^k(t)})^T = \dot{\mathbf{e}}^k(t)$, in which $\mathbf{e}^k(t) = \mathbf{x}^k(t) - \mathbf{x}_d(t)$ is a position tracking error at the k -th iteration. According to (5), one gets $\mathbf{v}^k(t) \rightarrow \mathbf{v}_d(t)$ to minimize $\mathcal{L}$. Thus, B_d^k and K_d^k can be iterated by

$$\begin{cases} B_d^k(t) = B_d^{k-1}(t) - \alpha \mathbf{e}^k(t)(\dot{\mathbf{v}}^k(t) - \dot{\mathbf{v}}_d(t))^T \\ K_d^k(t) = K_d^{k-1}(t) - \alpha \mathbf{e}^k(t)(\mathbf{v}^k(t) - \mathbf{v}_d(t))^T \end{cases} \quad (9)$$

in which $\alpha \in \mathbb{R}^+$ is a learning rate that absorbs β and α' . In summary, the impedance parameters B_d and K_d can be iterated by (9) to minimize $\mathcal{L}$, but the impedance IL needs to store the learned B_d and K_d at every instant of the previous iteration and is applicable only for a single task [7].

Remark 1: Different interaction effects can be achieved by setting a proper matrix C in (4) to change the expected output $\mathbf{v}$. When the elements corresponding to the position $\mathbf{x}$ in C are set larger, robots are in pursuit of position tracking, and the force tracking performance may be degraded. In scenarios focusing on compliant interaction, the elements corresponding to the integrated force $(\int_0^t \mathbf{f}_e(\zeta) d\zeta)$ in C can be chosen larger and the position tracking error may increase. Note that we can set $C = [C_x, \mathbf{0}, C_f]$ to avoid the usage of the acceleration $\ddot{\mathbf{x}}(t)$ in (9) and guarantee the non-singularity of CB , where $C_x, C_d \in \mathbb{R}^{6 \times 6}$ are some positive-definite matrices.

Remark 2: It is reasonable that the unknown environment is described by a mass-damping-spring model in the Cartesian space as in (3), where the inertia matrix $M_e(t)$, damping matrix $B_e(t)$, and stiffness matrix $K_e(t)$ can be adjusted to represent different environments, such as the human limb dynamics in physical human-robot interaction and the viscoelastic object dynamics in robotic manipulation [6].

Remark 3: The desired inertia matrix M_d in (6) is not

updated in this study since arbitrarily setting M_d may cause instability [29]. Actually, if M_d is reduced below its physical value, the passivity of the closed-loop system cannot be guaranteed due to the presence of unavoidable structural compliance between joint actuators and the interaction force, where passivity is a sufficient condition that guarantees coupled stability of robots interacting with passive environments [30].

B. Impedance Learning With Gaussian Process

The basic principle of the proposed IL-SOGP impedance learning method is shown in Fig. 1. In this study, the position $\mathbf{x}$, velocity $\dot{\mathbf{x}}$, and interaction force $\mathbf{f}_e$ at the robot end-effector, as well as the desired position $\mathbf{x}_d$ and velocity $\dot{\mathbf{x}}_d$, are applied to estimate the impedance parameters B and K [22], where $\mathbf{x}$ and $\mathbf{f}_e$ can be measured by an inertial measuring unit and a force sensor, respectively. To enhance the generalization of impedance learning and reduce the space complexity of data storage, define a variable impedance strategy

$$\begin{cases} B_d(t) = \pi_{B_d}(\mathbf{x}(t), \dot{\mathbf{x}}(t), \mathbf{x}_d(t), \dot{\mathbf{x}}_d(t), \mathbf{f}_e(t)) \\ K_d(t) = \pi_{K_d}(\mathbf{x}(t), \dot{\mathbf{x}}(t), \mathbf{x}_d(t), \dot{\mathbf{x}}_d(t), \mathbf{f}_e(t)) \end{cases} \quad (10)$$

where π_{B_d} and π_{K_d} are represented by iterative GPs [31]

$$\pi_{B_d} \sim \mathcal{GP}^k(0, \kappa(\mathbf{z}, \mathbf{z}')), \pi_{K_d} \sim \mathcal{GP}^k(0, \kappa(\mathbf{z}, \mathbf{z}')) \quad (11)$$

with $\mathbf{z}, \mathbf{z}' \in \mathcal{Z} = \{(\mathbf{x}(t), \dot{\mathbf{x}}(t), \mathbf{x}_d(t), \dot{\mathbf{x}}_d(t), \mathbf{f}_e(t)) | t \geq 0\}$, in which $\mathcal{GP}^k$ denotes the GP at the k -th iteration, and $\kappa(\mathbf{z}, \mathbf{z}') : \mathbb{R}^{18} \times \mathbb{R}^{18} \rightarrow \mathbb{R}$ is a Gaussian kernel function that represents a measure of the correlation of two points $(\mathbf{z}, \mathbf{z}')$. Online learning is required for continuous updates in (11). Online GPs can be divided into sparse approximation GPs and local GPs [11], in which the former is more applicable to massive data, and the latter requires considering smoothness among multiple models. We employ an SOGP because it can provide stable prediction while keeping a small size of the representative dataset. Later, this set will be called a basis vector ($\mathcal{BV}$) set.

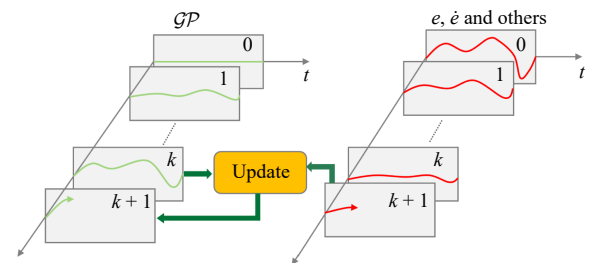


Fig. 1. The basic principle of the proposed IL-SOGP impedance learning method. Note that the proposed method models a variable impedance strategy with an iterative update of the SOGP, where the update part includes the task level (k -iteration) and the time level (t -axis).

SOGP keeps the $\mathcal{BV}$ set based on the metric of data points in the reproducing kernel Hilbert space (RKHS) by adding and deleting data schemes. In the update of a SOGP, for a new arrival point $\mathbf{z}_* \in \mathcal{Z}$, the residual error γ_{z_*} is given by [12]

$$\gamma_{z_*} = \mathcal{K}_{z_* z_*} - \mathcal{K}_{z_* Z} \mathcal{K}_{ZZ}^{-1} \mathcal{K}_{Z z_*}^T \quad (12)$$

where Z collects all points in the $\mathcal{BV}$ set, and $\mathcal{K}_{ZZ}$ denotes a covariance matrix. Note that γ_{z_*} can be seen as the “novelty”

measure of z_* for the $\mathcal{BV}$ set. If γ_{z_*} is greater than a threshold $\epsilon \in \mathbb{R}^+$, z_* is added to the $\mathcal{BV}$ set. Then, if the size of the $\mathcal{BV}$ set is excessive, a data point is selected for deletion based on an RKHS-based similar metric. Compared to the classical GP that requires inverting the covariance matrix of the order $O(N^3)$, the complexity of the proposed SOGP-IL can be reduced to the order $O(NM^2)$, where N is the number of training data, $M \ll N$ is the number of support points [32]. Thus, the proposed SOGP-IL has lower complexity.

It is not feasible to use SOGPs directly for modeling the variable impedance strategy in (10). With the continuous iterations of the task, the SOGP will not be updated for similar inputs. Hence, a pre-deletion operation can be added at the beginning of the update process to delete the points that are similar to the previous iterations. We introduce a hyperparameter $w \in [0, 1]$ as the threshold of similarity and delete the points whose similarity exceeds the threshold w in the $\mathcal{BV}$ set. Let z_i^{k-1} be a data point in the $\mathcal{BV}$ set before the previous $(k-1)$ -th iteration and $z^k \in \mathcal{Z}$ be a new arrival point at the current k -th iteration, where $i = 1$ to N , and N denotes the size of the $\mathcal{BV}$ set. Then, the similarity between the two data points is measured as follows:

$$d_i = \kappa(z_i^{k-1}, z^k), \quad i = 1, \dots, N \quad (13)$$

where the data points z_i^{k-1} with the similarity $d_i > w$ on the new arrival point z^k is deleted before each update of the SOGP. The detailed algorithm is shown in Algorithm 1.

Algorithm 1 Updating IL-SOGP in the k -th Iteration

Require: z^k , $\mathcal{BV}$ set, ϵ
 Compute $d_i = \kappa(z_i^{k-1}, z^k)$ by (13)
 Delete the corresponding point in $\mathcal{BV}$ set with $d_i > w$
 Compute the residual error γ_{z^k} by (12)
if $\gamma_{z^k} > \epsilon$ **then**
 Add the new arrival point z^k to $\mathcal{BV}$ set
else
 Only update the corresponding variables of SOGP
end if
if $\text{size}(\mathcal{BV}) > N$ **then**
 Delete a point based on RKHS-based scheme [12]
end if

Let t denote the “current” instant at any iteration cycle. If w is set too small, it causes the deletion of many data points in the $\mathcal{BV}$ set from the $(k-1)$ -th iteration. Although this allows learning new knowledge rapidly, it leads to a lack of data points in the $\mathcal{BV}$ set beyond the instant t of the $(k-1)$ -th iteration, resulting in possibly less accurate prediction. If w is set too large, few data points will be deleted, which leads to a lack of data points from the current k -th iteration. It means that the SOGP has difficulty in learning new knowledge, so the predicted values of the SOGP converge to those in the $(k-1)$ -th iteration. The setting of w is related to the hyperparameters of the kernel function κ , which can be set to a value where some newly arriving data points can be inserted into the $\mathcal{BV}$ set. With the exploitation of both the new knowledge of the current k -th iteration and the knowledge learned from the

previous $k-1$ iterations, the proposed IL-SOGP is expected to have better convergence than the original IL method.

C. Impedance Convergence Analysis

This section analyzes the convergence of the impedance parameters B_d and K_d in (10) iterated by (9).

Theorem 1: Consider the convex and differentiable objective function $\mathcal{L}$ in (7), where it is gradient Lipschitz continuous with a constant $L \in \mathbb{R}^+$. If the impedance parameters B_d^k and K_d^k are iterated by (9), i.e., $z^k = z^{k-1} - \beta \nabla \mathcal{L}(z^{k-1})$ with $z^k = [B_{d,11}^k, \dots, B_{d,66}^k, K_{d,11}^k, \dots, K_{d,66}^k]^T \in \mathbb{R}^{12}$, there exists an optimal solution z^* such that one has $\lim_{k \rightarrow \infty} \mathcal{L}(z^k) = \mathcal{L}(z^*)$, and the gradient $\nabla \mathcal{L}(z^*)$ satisfies $\|\nabla \mathcal{L}(z^*)\|_2^2 = 0$, where $\mathcal{L}(z^*)$ denotes the optimal value of $\mathcal{L}$.

Proof: Applying a quadratic expansion to the objective function $\mathcal{L}(z^k)$ around $\mathcal{L}(z^{k-1})$, one obtains

$$\begin{aligned} \mathcal{L}(z^k) &\leq \mathcal{L}(z^{k-1}) + \nabla \mathcal{L}(z^{k-1})^T (z^k - z^{k-1}) \\ &\quad + \frac{1}{2} (z^k - z^{k-1})^T \nabla^2 \mathcal{L}(z^{k-1}) (z^k - z^{k-1}). \end{aligned}$$

Since $\nabla \mathcal{L}$ is Lipschitz continuous with the constant L , i.e., $\nabla^2 \mathcal{L}(x) - LI < 0$, $\forall x \in \mathbb{R}^{12}$, one obtains

$$\begin{aligned} \mathcal{L}(z^k) &\leq \mathcal{L}(z^{k-1}) + \nabla \mathcal{L}(z^{k-1})^T (z^k - z^{k-1}) \\ &\quad + \frac{1}{2} L \|z^k - z^{k-1}\|_2^2. \end{aligned}$$

Applying $z^k = z^{k-1} - \beta \nabla \mathcal{L}(z^{k-1})$ to the above result yields

$$\begin{aligned} \mathcal{L}(z^k) &\leq \mathcal{L}(z^{k-1}) - \beta \|\nabla \mathcal{L}(z^{k-1})\|_2^2 + \frac{1}{2} L \beta^2 \|\nabla \mathcal{L}(z^{k-1})\|_2^2 \\ &= \mathcal{L}(z^{k-1}) - (1 - \frac{1}{2} \beta L) \beta \|\nabla \mathcal{L}(z^{k-1})\|_2^2. \end{aligned}$$

Choosing $\beta < 1/L$, one obtains

$$\mathcal{L}(z^k) \leq \mathcal{L}(z^{k-1}) - \frac{1}{2} \beta \|\nabla \mathcal{L}(z^{k-1})\|_2^2$$

which implies that $\mathcal{L}(z^k)$ strictly decreases with each iteration of gradient descent until $\|\nabla \mathcal{L}(z^*)\|_2^2 = 0$ and $\lim_{k \rightarrow \infty} \mathcal{L}(z^k) = \mathcal{L}(z^*)$, where $\mathcal{L}(z^*)$ is the optimal value of $\mathcal{L}$. ■

D. Discussions on Pros and Cons

Compared to parameterized machine-learning models such as NNs, SOGP only needs to set a small number of hyperparameters (rather than to training a large number of weight parameters), and the size of parameters does not increase with the increase of the number of training data. Using the existing data, SOGP can give the predicted value of the new data with its uncertainty through Bayesian inference, which can be explained by probability theory. SOGP pays more attention to interpretation and explores the underlying pattern of data through kernel functions, whereas NN pays more attention to generalization and discovers the underlying patterns of data at the expense of interpretability and training efficiency. Therefore, compared to SOGP, NN requires a long training time with a larger amount of training data and cannot respond to

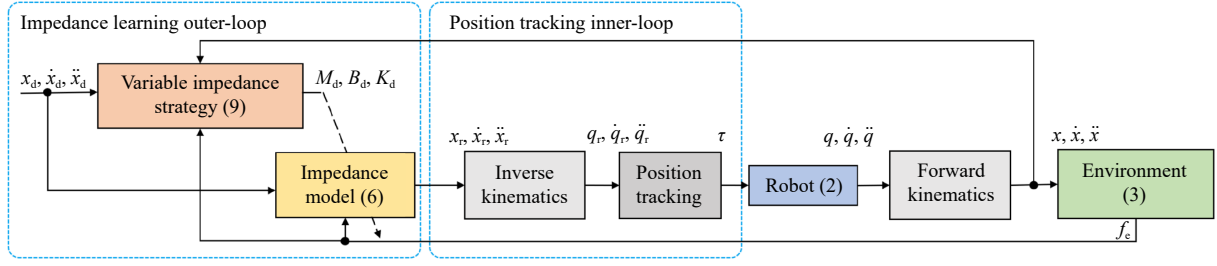


Fig. 2. A variable impedance control framework based on robot admittance control. Note that the variable impedance strategy provides the proper impedance parameters, and then the impedance model generates a reference trajectory for robot tracking.

real-time changes quickly.

The proposed IL-SOGP method for varying target impedance follows the impedance IL framework in [6] that does not consider the stability of impedance learning as in most optimization-based (or learning-based) variable impedance methods [4]. This is in contrast to adaptive variable impedance methods that can usually guarantee stable impedance adaptation based on the Lyapunov synthesis [33], [34]. Stability in impedance variation is always important when robots physically interact with environments. The proposed method minimizes both the tracking error e and the interaction force f_e in the objective function $\mathcal{L}$ such that the impedance parameters B_d and K_d in (9) tend to achieve stable and compliant interaction, but this does not have a theoretical guarantee, and the stability problem is still open. One possible solution is to impose a strict relationship between B_d and K_d to always have a critically-damped response of robot control. Another possible solution is to impose a Lyapunov stability guarantee on the robot's final behavior, but analyzing interaction stability with Lyapunov-based tools becomes difficult when the environmental dynamics is unknown [35].

Other limitations of the proposed IL-SOGP method include: 1) In the inner control loop, it is assumed that an accurate robot model is known such that the reference trajectory x_r can be tracked accurately, which may be alleviated by the combination of model-free GP control methods such as [36]; 2) The objective function $\mathcal{L}$ in (7) considers tracking the integrated interaction force $\int_0^t f_e(\varsigma) d\varsigma$ instead of the interaction force f_e , which limits its application scope; 3) Hardware experiments are challenging due to the high computational cost of SOGP [32], where the complexity of SOGP increases as the increase of the number of training data such that the real-time performance is difficult to guarantee in practice.

IV. NUMERICAL VERIFICATION

A. Simulation Environment Setup

This section validates the convergence and generalization of the proposed IL-SOGP method based on a simulated 7-DoF collaborative robot called Franka Emika Panda in the MATLAB software, where the dynamic parameters of the Panda robot are from the high-fidelity identification result in [37]. The inner loop implements a computed torque position controller with a sampling delay of 1 ms, and the outer-loop sampling delay for admittance control is 10 ms. For clear illustrations, we only consider variable impedance for the z -axis translation. The control framework is depicted in Fig. 2, and

more details on the robot kinematics can be seen in [23].

The environment (3) is to be an uneven object with a height $h = 0.5 + 0.003 \sin(80 x_z(t) + 1.2) + \varepsilon_n$, where $x_z(t) \in \mathbb{R}$ is the z -direction of $x(t)$, $\varepsilon_n \sim \mathcal{N}(0, 0.0015^2)$ is Gaussian noise with a mean 0 and a variance 0.0015^2 , $M_e = 0.01\Phi(t)I$, $B_e = 0.1\Phi(t)I$, and $K_e = 300\Phi(t)I$ with $\Phi(t) = (0.5 \sin(2\pi t) + 0.5)^2$ [6]. The initial position of the end-effector is $x_0 = [0.555, 0, 0.515]^T$ m, the desired interaction force is $f_d = 0$ N, and the desired motion is the end-effector moving downward the object and sliding a distance after contact with its surface.

The damping parameter B_d and stiffness parameter K_d in the z -axis direction are modeled by two GPs as in (11), and are initialized as 100 and 20, respectively. Choose a squared exponential kernel $\kappa(z, z') = \sigma_s^2 \exp(-(z - z')^T \Lambda (z - z')/2)$ and set $w = 0.8$, $\epsilon = 0.001$, $N = 15$, $\sigma_s = 1$, $\sigma_n = 0.001$, and $\Lambda = \text{diag}\{3, 3, 3, 3, 4\}$ for both GPs, where $\sigma_s^2 \in \mathbb{R}^+$ is a signal variance, and $\Lambda = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_5\}$ with $\lambda_i \in \mathbb{R}^+$ ($i = 1$ to 5) is a length-scale matrix of the input vector [31]. Note that in the following figures, red curves are generated by the classical IL, blue curves are generated by the proposed IL-SOGP, orange curves denote the desired trajectories, deeper color denotes higher iterative times, and green arrows indicate the evolving direction of iterations from $k = 1$ to 5 or 10.

B. Comparison of Convergence Performance

The convergence performance of the classical IL method in [6] and the proposed IL-SOGP method are explored under two cases: Case 1 for accurate tracking with $\alpha = 10$ and $C = [25, 0, 1]$ and Case 2 for compliant interaction with $\alpha = 20$ and $C = [100, 0, 0.1]$. From Figs. 3 and 4, it is clear that in Case 1, the damping parameter B_d and the stiffness parameter K_d become larger for better position tracking, and in Case 2, B_d and K_d become smaller, resulting in a smaller interaction force f_e for compliant interaction. However, the convergence rate of the IL-SOGP is higher than for both B_d and K_d because it exploits the new iteration values learned from the previous instants in the same iteration, which provides a trend for the subsequent prediction and thus accelerates convergence. Also, the IL-SOGP exhibits more sharp changes of B_d and K_d , which is caused by the low frequency of the admittance control part and results in a sharp acceleration when data points are added to the $\mathcal{BV}$ set. But for the variable impedance control, these low-frequency changes are subtle, and no sudden changes in the interaction force f_e or the robot position x will be introduced during the interaction. The average of the objective function $\mathcal{L}$ in (7) by the IL-SOGP converges much faster, as exhibited in Figs. 3(a) and 3(b). The

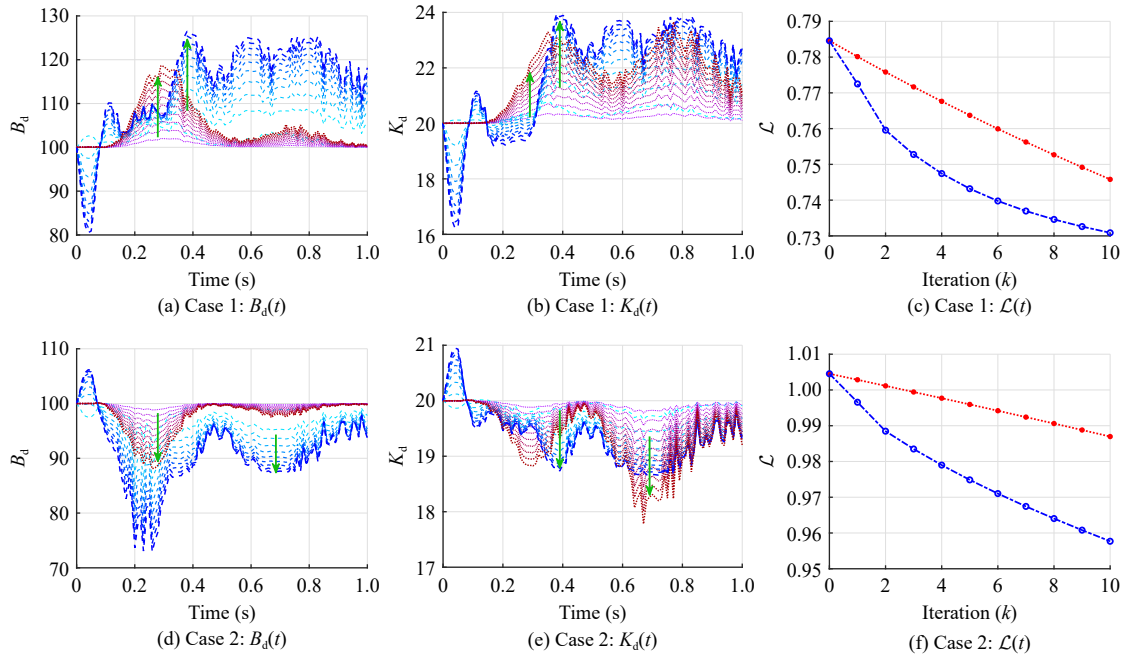


Fig. 3. Convergence results by the two learning methods in the disturbance-free scenario of Section IV-B ($k = 10$). Note that red curves are by the classical IL, blue curves are by the proposed IL-SOGP, and arrows indicate the evolving direction of the iterations from $k = 1$ to 10.

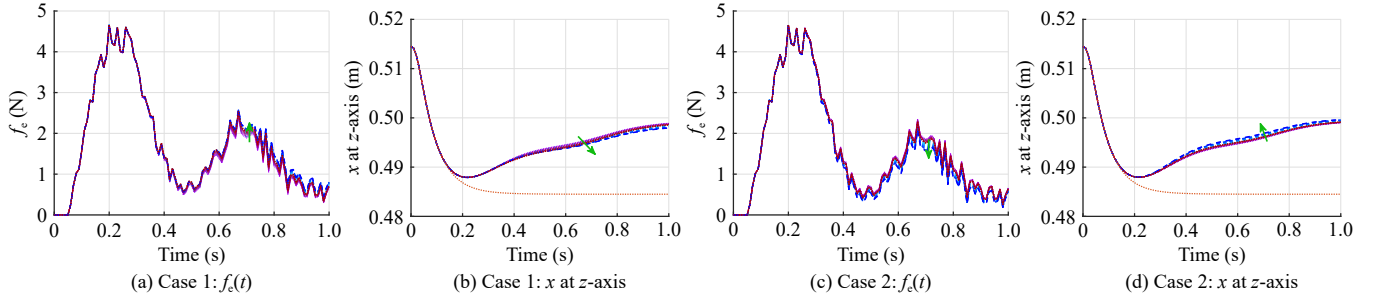


Fig. 4. Interaction processes by the two learning methods in the disturbance-free scenario of Section IV-B ($k = 10$). Note that red curves are by the classical IL, blue curves are by the proposed IL-SOGP, orange curves are the desired trajectories, and arrows indicate the evolving direction of the iterations from $k = 1$ to 10.

predicted values of $\mathcal{L}$ by the two methods are exhibited in Tables I and II, respectively, where $\mathcal{L}$ can also represent a prediction error to reflect the estimation accuracy of B_d^k and K_d^k in (10). It is shown that the IL-SOGP provides smaller mean values and variances, and after 3 iterations, it performs better than the classical IL after 10 iterations. In summary, compared to the classical IL, the proposed IL-SOGP has better position tracking and faster iterations in Case 1 and has a smaller interaction force f_e , resulting in more compliant interaction in Case 2.

C. Comparison of Generalization Performance

The classical IL method requires re-iteration when the task or environment is changed, which restricts its applications to scenarios interacting with unknown environments. To verify the generalization of the proposed IL-SOGP method, we denote Case 1 in Section IV-B as Task 1, and introduce extra Tasks 2 and 3 with modified initial positions and desired trajectories (As shown in Figs. 5 and 6), where the initial position $\mathbf{x}_0$ in the z -axis is tuned to 0.503 m and the desired trajec-

tory $\mathbf{x}_d$ is designed as in Fig. 6(a) for Tasks 2 and 3 (orange curves). For Task 2, set $\Phi(t) = (0.7\sin(2\pi t) + 0.3)^2$ such that the parameter variation is more obvious. For Task 3, set $h = 0.5 + 0.006\sin(80x_c(t) + 1.2) + \varepsilon_n$ such that the environment is more undulating. It is worth noting that the change of $\mathbf{x}_0$ makes the robot contact the environment earlier.

It is observed from Fig. 5 that the proposed IL-SOGP keeps small changes for the five iterations of the damping parameter B_d and the stiffness parameter K_d , while the changes of B_d and K_d by the classical IL are large because of restarting iterations. From Figs. 5(a) and 5(b), one knows that the iterative changes of the averaged $\mathcal{L}$ by the IL-SOGP are small, indicating that it is close to the final convergence. The above results validate that the IL-SOGP can exploit the learned knowledge reasonably well in the case of switching tasks, resulting in fast convergence to provide a new variable impedance strategy suitable for the new task. The interaction process in Fig. 6 also shows that the IL-SOGP has better position tracking and is close to the final convergence. In summary, the proposed IL-SOGP is able to predict proper impedance parameters for the new task based on the learned variable impedance strategy.

TABLE I
MEAN VALUES OF THE OBJECTIVE FUNCTION $\mathcal{L}$ AT EACH ITERATION

Iteration	1	2	3	4	5	6	7	8	9	10
Case 1 (Classical IL)	0.780	0.776	0.772	0.768	0.764	0.760	0.756	0.753	0.749	0.749
Case 1 (IL-SOGP)	0.773	0.760	0.753	0.747	0.743	0.740	0.737	0.735	0.733	0.731
Case 2 (Classical IL)	1.003	1.001	0.999	0.998	0.996	0.994	0.992	0.991	0.989	0.987
Case 2 (IL-SOGP)	0.997	0.989	0.984	0.980	0.975	0.971	0.967	0.964	0.961	0.958

TABLE II
VARIANCES OF THE OBJECTIVE FUNCTION $\mathcal{L}$ AT EACH ITERATION

Iteration	1	2	3	4	5	6	7	8	9	10
Case 1 (Classical IL)	0.256	0.254	0.251	0.249	0.247	0.244	0.242	0.240	0.238	0.236
Case 1 (IL-SOGP)	0.251	0.241	0.236	0.232	0.229	0.226	0.224	0.222	0.221	0.220
Case 2 (Classical IL)	0.315	0.313	0.312	0.310	0.308	0.306	0.305	0.303	0.301	0.300
Case 2 (IL-SOGP)	0.301	0.301	0.296	0.292	0.288	0.284	0.281	0.278	0.275	0.272

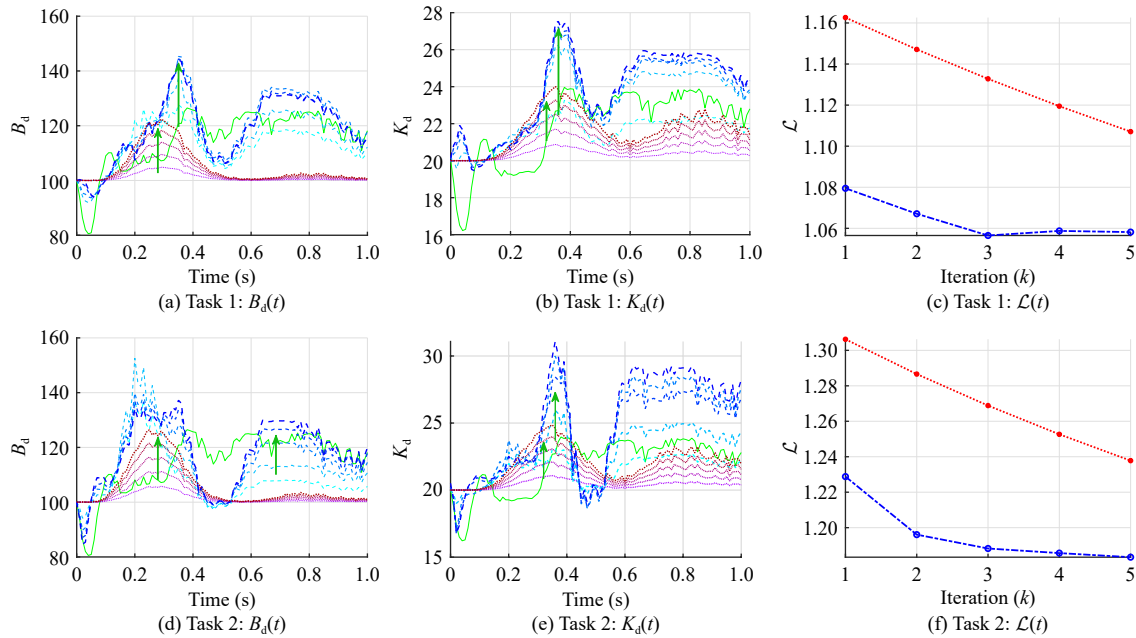


Fig. 5. Generalization results by the two learning methods in Tasks 1 and 2 of Section IV-C ($k = 5$). Note that red curves are by the classical IL, blue curves are by the proposed IL-SOGP, green curves are from the 10th iteration in Task 1, and arrows indicate the evolving direction of the iterations from $k = 1$ to 5.

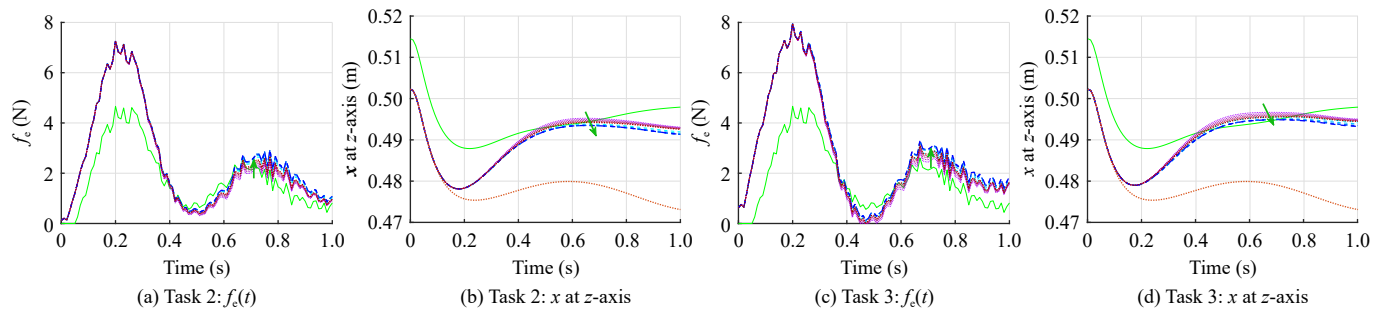


Fig. 6. Interaction processes by the two learning methods in Tasks 2 and 3 of Section IV-C ($k = 5$). Note that red curves are by the classical IL, blue curves are by the proposed IL-SOGP, green curves are from the 10th iteration in Task 1, orange curves are the desired trajectories, and arrows indicate the evolving direction of the iterations from $k = 1$ to 5.

D. Comparison of Robustness Performance

To demonstrate the robustness of the proposed IL-SOGP,

we choose the same cases and parameters as those in Section IV-B, except that the centripetal-Coriolis and friction compensation terms in the inner loop torque controller are igno-

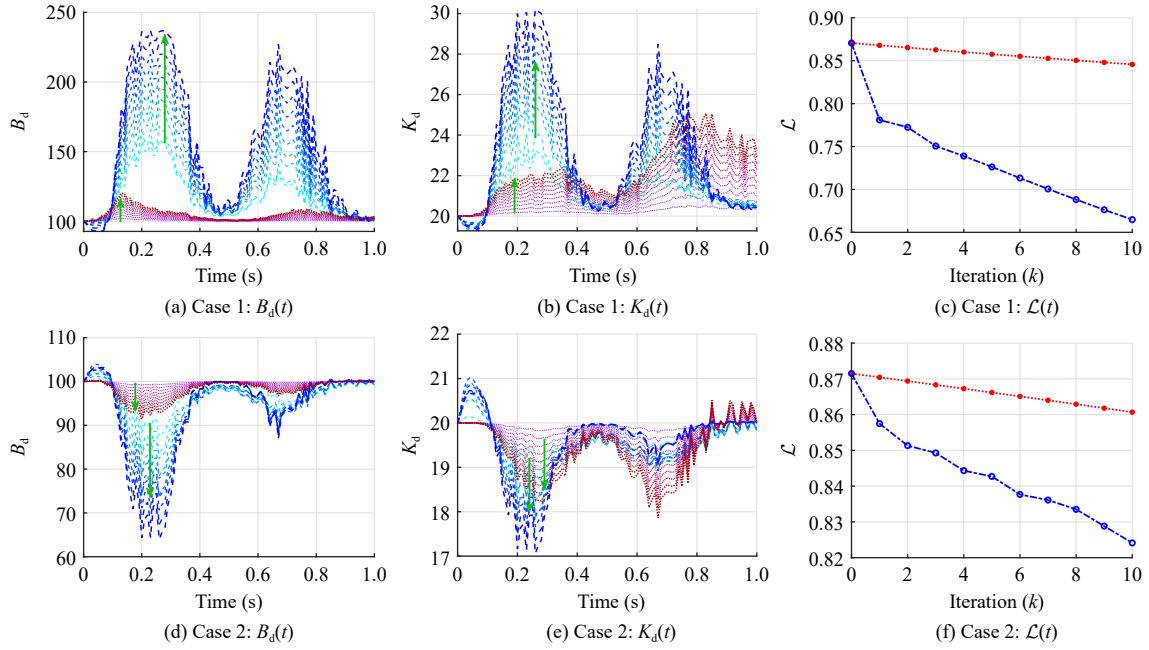


Fig. 7. Convergence results by the two learning methods in the disturbed scenario of Section IV-D ($k = 10$). Note that red curves are by the classical IL, blue curves are by the proposed IL-SOGP, and arrows indicate the evolving direction of the iterations from $k = 1$ to 10.

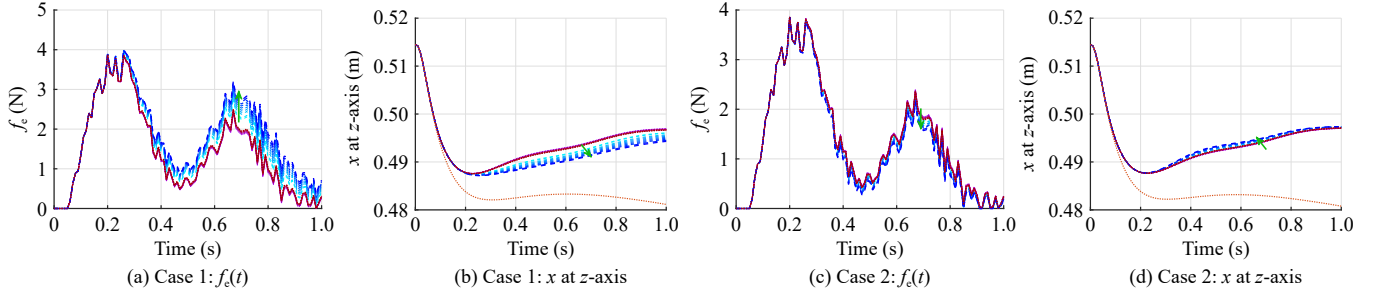


Fig. 8. Interaction processes by the two learning methods in the disturbed scenario of Section IV-D ($k = 10$). Note that red curves are by the classical IL, blue curves are by the proposed IL-SOGP, and arrows indicate the evolving direction of the iterations from $k = 1$ to 10.

red. An external disturbance $\mathbf{d}(t) = [0.4, 0.4, 0.4, 0.4, 0.2, 0.2, 0.2]^T \sin(\pi/2t)$ is added to the right side of the robot dynamics (2). In this scenario, the parameter uncertainties from centripetal-Coriolis and friction forces, external disturbance $\mathbf{d}$, and measurement noise affect robot interaction. Simulation results are shown in Figs. 7 and 8. Compared to the disturbance-free scenario in Section IV-B, the changes of the damping parameter B_d and stiffness parameter K_d by the proposed IL-SOGP are larger owing to the parameter uncertainties and the external disturbance $\mathbf{d}$. In Case 1, B_d and K_d by the IL-SOGP become larger for better position tracking, and in Case 2, B_d and K_d become smaller for compliant interaction. The IL-SOGP owns a higher convergence rate for both B_d and K_d because it exploits the new iteration values learned from the previous instants in the same iteration, which provides a trend for the subsequent prediction and thus accelerates convergence. The averaged $\mathcal{L}$ by the IL-SOGP shows faster convergence, which implies that it is robust against parameter uncertainties and external disturbances.

V. CONCLUSIONS AND FUTURE WORK

In this paper, an IL-SOGP impedance learning framework has been proposed to improve learning convergence and gen-

eralization during robot-environment interaction. Numerical verification has shown that compared with the classical impedance IL method, the proposed method provides better convergence and generalization in modeling the variable impedance strategy for a 7-DoF robot arm. In future work, we would consider novel objective functions, adaptive similarity, advanced inner-loop control [38]–[40], interaction stability analysis under unknown environments, interaction with more complex environments, and physical experiments based on more efficient GP models.

REFERENCES

- [1] E. Matheson, R. Minto, E. G. Zampieri, M. Faccio, and G. Rosati, "Human-robot collaboration in manufacturing applications: A review," *Robot.*, vol. 8, no. 4, p. 100, Dec. 2019.
- [2] G. Kang, H. S. Oh, J. K. Seo, U. Kim, and H. R. Choi, "Variable admittance control of robot manipulators based on human intention," *IEEE/ASME Trans. Mechatronics*, vol. 24, no. 3, pp. 1023–1032, Jun. 2019.
- [3] C. Ott, R. Mukherjee, and Y. Nakamura, "A hybrid system framework for unified impedance and admittance control," *J. Intell. Robot. Syst.*, vol. 78, no. 3–4, pp. 359–375, Jul. 2015.
- [4] M. Sharifi, A. Zakerimaneh, J. K. Mehr, A. Torabi, V. K. Mushahwar, and M. Tavakoli, "Impedance variation and learning strategies in

- human-robot interaction,” *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 6462–6475, Jul. 2022.
- [5] N. Nikooinnejad, M. Maroufi, and S. O. Reza Moheimani, “A survey of iterative learning control: A learning-based method for high-performance tracking control,” *IEEE Control Syst. Mag.*, vol. 26, no. 3, pp. 96–114, Jun. 2006.
 - [6] Y. Li and S. S. Ge, “Impedance learning for robots interacting with unknown environments,” *IEEE Trans. Control Syst. Technol.*, vol. 22, no. 4, pp. 1422–1432, Jul. 2014.
 - [7] S. Riaz, L. Hui, M. S. Aldemir, and F. Afzal, “A future concern of iterative learning control: A survey,” *J. Stat. Manag. Syst.*, vol. 24, no. 6, pp. 1301–1322, Apr. 2021.
 - [8] D. Zhang, Z. Wang, and T. Masayoshi, “Neural-network-based iterative learning control for multiple tasks,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 9, pp. 4178–4190, Sept. 2020.
 - [9] L. Rath, A. R. Geist, and S. Trimpe, “Using physics knowledge for learning rigid-body forward dynamics with Gaussian process force priors,” in *Proc. Conf. Robot Learn.*, London, UK, 2021, pp. 101–111.
 - [10] J. Lee, J. Feng, M. Humt, M. G. Müller, and R. Triebel, “Trust your robots! predictive uncertainty estimation of neural networks with sparse Gaussian processes,” in *Proc. Conf. Robot Learn.*, London, UK, 2021, pp. 1168–1179.
 - [11] H. Liu, Y.-S. Ong, X. Shen, and J. Cai, “When Gaussian process meets big data: A review of scalable GPs,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 11, pp. 4405–4423, Nov. 2020.
 - [12] L. Csato and M. Opper, “Sparse on-line Gaussian processes,” *Neural Comput.*, vol. 14, no. 3, pp. 641–668, Mar. 2002.
 - [13] J. S. De La Cruz, W. Owen, and D. Kulic, “Online learning of inverse dynamics via Gaussian process regression,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, Vilamoura, Algarve, Portugal, 2012, pp. 3583–3590.
 - [14] L. Deng, W. Li, and Y. Pan, “Data-efficient Gaussian process online learning for adaptive control of multi-DoF robotic arms,” *IFAC-PapersOnLine*, vol. 55, no. 2, pp. 84–89, 2022.
 - [15] R.-E. Precup, S. Preitl, J. K. Tar, M. L. Tomescu, M. Takacs, P. Korondi, and P. Baranyi, “Fuzzy control system performance enhancement by iterative learning control,” *IEEE Trans. Ind. Electron.*, vol. 55, no. 9, pp. 3461–3475, Sept. 2008.
 - [16] J. Wang, Y. Gao, Y. Gao, J. Liu, G. Sun, and L. Wu, “Intelligent dynamic practical-sliding-mode control for singular Markovian jump systems,” *Inf. Sci.*, vol. 607, pp. 153–172, Aug. 2022.
 - [17] I. A. Zamfirache, R.-E. Precup, R.-C. Roman, and E. M. Petriu, “Neural network-based control using actor-critic reinforcement learning and grey wolf optimizer with experimental servo system validation,” *Expert Syst. Appl.*, vol. 225, Art. no. 120112, Sept. 2023.
 - [18] F. Dimeas and N. Aspragathos, “Fuzzy learning variable admittance control for human-robot cooperation,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, Chicago, IL, USA, 2014, pp. 4770–4775.
 - [19] A.-N. Sharkawy, P. N. Koustournparris, and N. Aspragathos, “Variable admittance control for human-robot collaboration based on online neural network training,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, Madrid, Spain, 2018, pp. 1334–1339.
 - [20] A.-N. Sharkawy, P. Koustournparris, and N. Aspragathos, “A neural network-based approach for variable admittance control in human-robot cooperation: Online adjustment of the virtual inertia,” *Intel. Serv. Robotics*, vol. 13, pp. 495–519, Aug. 2020.
 - [21] X. Yu, W. He, Y. Li, C. Xue, J. Li, J. Zou, and C. Yang, “Bayesian estimation of human impedance and motion intention for human-robot collaboration,” *IEEE Trans. Cybern.*, vol. 51, no. 4, pp. 1822–1834, Apr. 2021.
 - [22] C. Li, Z. Zhang, G. Xia, X. Xie, and Q. Zhu, “Efficient learning variable impedance control for industrial robots,” *Bull. Pol. Acad. Sci. Tech. Sci.*, vol. 67, p. 2, Apr. 2019.
 - [23] L. Deng, Z. Li, and Y. Pan, “Sparse online Gaussian process impedance learning for multi-DoF robotic arms,” in *Proc. IEEE Int. Conf. Adv. Robot. Mechatron.*, Chongqing, China, 2021, pp. 199–206.
 - [24] C. Wang, Y. Li, S. S. Ge, and T. H. Lee, “Reference adaptation for robots in physical interactions with unknown environments,” *IEEE Trans. Cybern.*, vol. 47, no. 11, pp. 3504–3515, Nov. 2017.
 - [25] T. Yamawaki, H. Ishikawa, and M. Yashima, “Iterative learning of variable impedance control for human-robot cooperation,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, Daejeon, South Korea, 2016, pp. 839–844.
 - [26] A. Kramberger, E. Shahriari, A. Gams, B. Nemec, A. Ude, and S. Haddadin, “Passivity based iterative learning of admittance-coupled dynamic movement primitives for interaction with changing environments,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, Madrid, Spain, 2018, pp. 6023–6028.
 - [27] X. Li, Y.-H. Liu, and H. Yu, “Iterative learning impedance control for rehabilitation robots driven by series elastic actuators,” *Automatica*, vol. 90, pp. 1–7, Apr. 2018.
 - [28] K. Dupree, C.-H. Liang, G. Hu, and W. E. Dixon, “Adaptive lyapunov-based control of a robot and mass-spring system undergoing an impact collision,” *IEEE Trans. Syst. Man Cybern. B*, vol. 38, no. 4, pp. 1050–1061, Aug. 2008.
 - [29] J. E. Colgate and N. Hogan, “Robust control of dynamically interacting systems,” *Int. J. Control*, vol. 48, no. 1, pp. 65–88, Oct. 1988.
 - [30] F. Ficuciello, L. Villani, and B. Siciliano, “Variable impedance control of redundant manipulators for intuitive human-robot physical interaction,” *IEEE Trans. Robot.*, vol. 31, no. 4, pp. 850–863, Aug. 2015.
 - [31] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. Cambridge, MA, USA: MIT press, 2006.
 - [32] B. Wilcox and M. C. Yip, “SOLAR-GP: Sparse online locally adaptive regression using Gaussian processes for Bayesian robot model learning and control,” *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 2832–2839, Apr. 2020.
 - [33] X. Li, Y. Pan, G. Chen, and H. Yu, “Adaptive human-robot interaction control for robots driven by series elastic actuators,” *IEEE Trans. Robot.*, vol. 33, no. 1, pp. 169–182, Feb. 2017.
 - [34] T. Sun, L. Peng, L. Cheng, Z.-G. Hou, and Y. Pan, “Stability-guaranteed variable impedance control of robots based on approximate dynamic inversion,” *IEEE Trans. Syst. Man, Cybern. Syst.*, vol. 51, no. 7, pp. 4193–4200, Jul. 2019.
 - [35] F. J. Abu-Dakka and M. Saveriano, “Variable impedance control and learning—A review,” *Front. Robot. AI*, vol. 7, p. 177, Dec. 2020.
 - [36] W. Li, Z. Li, Y. Liu, and Y. Pan, “Learning inverse robot dynamics using sparse online Gaussian process with forgetting mechanism,” in *Proc. IEEE Int. Conf. Adv. Intell. Mechatronics*, Sapporo, Japan, 2022, pp. 638–643.
 - [37] C. Gaz, M. Cognetti, A. Oliva, P. R. Giordano, and A. De Luca, “Dynamic identification of the Franka Emika Panda robot with retrieval of feasible parameters using penalty-based optimization,” *IEEE Robot. Autom. Lett.*, vol. 4, no. 4, pp. 4147–4154, Oct. 2019.
 - [38] Y. Xie, X. Zhang, S. Zheng, C. K. Ahn, and S. Wang, “Asynchronous H_∞ continuous stabilization of mode-dependent switched mobile robot,” *IEEE Trans. Syst. Man, Cybern. Syst.*, vol. 52, pp. 6906–6920, Nov. 2022.
 - [39] S.-K. Kim, C. K. Ahn, and R. K. Agarwal, “Position-tracking controller for two-wheeled balancing robot applications using invariant dynamic surface,” *IEEE Trans. Syst. Man, Cybern. Syst.*, vol. 51, no. 2, pp. 705–711, Feb. 2021.
 - [40] M.-G. Unguritu and T.-C. Nichiștea, “Design and assessment of an anti-lock braking system controller,” *Roman. J. Inf. Sci. Technol.*, vol. 26, no. 1, pp. 21–32, 2023.



Yongping Pan (Senior Member, IEEE) received the Ph.D. degree in control theory and control engineering from South China University of Technology in 2011.

He spent one year as a control systems engineer in the industry in Shenzhen and Guangzhou and has over ten years of research experience with the Nanyang Technological University, Singapore, the National University of Singapore, Singapore, and the University of Tokyo, Japan. In 2019, he joined the Sun Yat-sen University in Guangzhou and Shenzhen as a Professor who led the Intelligent Robotics Lab. He is currently a Professor with Southeast Uni-

versity. His research interests include automatic control and machine learning for robotics, such as compliant actuation, interaction control, visual servoing, motion planning, and dexterous manipulation. He has authored or co-authored over 200 peer-reviewed academic papers, including more than 140 papers in refereed journals.

Dr. Pan serves as the Chair of the IEEE Robotics and Automation Society Guangzhou Chapter, as a Specialty Chief Editor of *Frontiers in Robotics and AI*, and as an Associate Editor of several top-tier journals, such as *IEEE Transactions Series*. He has held various organizing roles in over 10 international conferences and served as Lead Guest Editor for four special issues in reputable refereed journals. Moreover, he received several best paper awards from major conferences in robotics and automation.



Tian Shi (Member, IEEE) received the M.S. degree in control engineering from Jilin University in 2021, and the Ph.D. degree in computer science and technology from Sun Yat-sen University in 2025.

His research interests include adaptive and interaction control for compliant actuation robots.



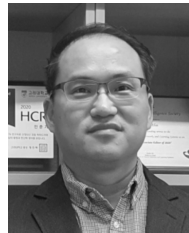
Wei Li received the B.S. degree in electronic information engineering from Guangdong University of Technology in 2017, and the M.S. degree in computer technology from Sun Yat-sen University in 2022.

His research interests include Gaussian process and physical robot interaction.



Bin Xu received the B.S. degree in measurement and control from Northwestern Polytechnical University in 2006, and the Ph.D. degree in computer science from Tsinghua University in 2012.

He is currently a Professor with the School of Automation, Northwestern Polytechnical University. His research interests include intelligent control and adaptive control with applications.



Choon Ki Ahn (Senior Member, IEEE) received the Ph.D. degree at the School of Electrical Engineering and Computer Science from Seoul National University, South Korea, in 2006.

He is currently a Distinguished Chair Professor with the School of Electrical Engineering, Korea University, South Korea. In 2017, he received the Presidential Young Scientist Award from the President of South Korea. In 2020, 2021, and 2023, he received the Best (Outstanding) Associate Editor

Award for *IEEE Transactions on Neural Networks and Learning Systems*; *IEEE Transactions on Systems, Man, and Cybernetics: Systems*; and *IEEE Transactions on Circuits and Systems I: Regular Papers*, respectively. In 2019–2025, he received the Korea University Research Excellence Award. In 2023, he was inducted into the Korea University Hall of Fame as the world's top researcher.

He has been a Senior Editor of *IEEE Transactions on Neural Networks and Learning Systems*; *IEEE Transactions on Systems, Man, and Cybernetics: Systems*; *IEEE Systems Journal*. He has also been an Associate Editor of *IEEE Transactions on Fuzzy Systems*; *IEEE Transactions on Automation Science and Engineering*; *IEEE Transactions on Intelligent Transportation Systems*; *IEEE Transactions on Circuits and Systems I: Regular Papers*; *IEEE Systems, Man, and Cybernetics Magazine*; and other flagship journals. He is the recipient of the 2019–2024 Highly Cited Researcher Award in Engineering by Clarivate Analytics (formerly, Thomson Reuters).